Kateri elementi kolobarja $\mathbb{Z}[i]$

so asociirani: $m+ni$

$$x = u(m+ni) = um + uni$$

kateri so obrnljivi:

$$(x+yi)(a+bi) = 1$$

$$xa - yb + i(ay+xb) = 1$$

$$ay + xb = 0 \qquad xa - yb = 1$$

$$x = \frac{-ay}{b} \qquad -\frac{a^2y}{b} - yb = 1$$

$$-\frac{y}{b}(a^2+b^2) = 1$$

$$y = \frac{-b}{a^2+b^2}$$

$$x = \frac{a}{a^2+b^2} \qquad |a| \leq |a^2| \Rightarrow$$

$$a^2+b^2 \in \{0,1\} \Rightarrow a^2+b^2 = 1$$

$$\Rightarrow a \in \{0,1\}$$
$$b \in \{0,1\} \qquad \text{obrnljivi} \quad \{1,-1,i,-i\}$$

asoc: $\{m+ni, -m-ni, n-mi, -n+mi$

$d \in \mathbb{Z}$

$\mathbb{Z}[\sqrt{d}] = \{m + n\sqrt{d};\ m, n \in \mathbb{Z}\}$

1) Pokaži $\mathbb{Z}[\sqrt{d}]$ je podkolobar $\mathbb{C}$

2) Množica $\mathbb{Q}[\sqrt{d}] = \{q + r\sqrt{d};\ q, r \in \mathbb{Q}\}$ je podpolje $\mathbb{C}$ generirano z $\mathbb{Z}[\sqrt{d}]$

---

1) $\mathbb{Z}[\sqrt{d}] \subseteq \mathbb{C}$

zaprtost za seštevanje, množenje, enota

za množenje:

$$(m+n\sqrt{d})(x+y\sqrt{d}) = mx + nyd + \sqrt{d}(ny + ym)$$

enota $= 1 = 1 + 0\cdot\sqrt{d}$

seštevanje: $(m+n\sqrt{d}) - (x+y\sqrt{d}) = (m-x) + (n-y)\sqrt{d}$

2) inverziranje

ja tu

$$\frac{m+n\sqrt{d}}{x+y\sqrt{d}} = \frac{(m+n\sqrt{d})(x-y\sqrt{d})}{x^2-y^2d} = \frac{c+e\sqrt{d}}{c^2+e^2d}$$

Avtomorfizmi $\mathbb{Z}[\sqrt{d}] = \{id;\ \underbrace{\sqrt{d} \mapsto -\sqrt{d}}_{\sigma}\}$

Norma $N(x) = x\sigma(x)$

$$N(q+r\sqrt{d}) = (q+r\sqrt{d})(q-r\sqrt{d}) = q^2 - r^2d$$

3) $\forall x,y \in \mathbb{Z}[\sqrt{a}]: N(xy) = N(x)N(y)$

$$N(xy) = xy\sigma(xy) = xy\sigma(x)\sigma(y) = x\sigma(x)y\sigma(y)$$
$$= N(x)\cdot N(y)$$

4) $x \in \mathbb{Z}[\sqrt{a}]$ je obrnljiv $\Leftrightarrow N(x) = \pm 1$

$$xy = 1 \Rightarrow N(xy) = N(1)$$
$$N(x)N(y) = 1$$
$$N(x) = \frac{1}{N(y)} \qquad N(y) \in \mathbb{Z} \wedge N(x) \in \mathbb{Z}$$
$$\Rightarrow N(y) \in \{-1, 1\}$$
$$\Rightarrow N(x) \in \{-1, 1\}$$

5) $p \in \mathbb{P}$; $N(x) = \pm p \Rightarrow x$ nerazcepen

in $N(x) = \pm p$
recimo da x razcepen ab nedrobljiva

$$x = ab$$
$$N(x) = N(a)N(b) = \pm p \Rightarrow N(a) = 1 \vee N(b) =$$

4) $\Leftarrow N(x) = 1$, $x\sigma(x) = 1 \Rightarrow x^{-1} = \sigma(x)$

5) nepr. ker je $N(x) = \pm 1 \Rightarrow a$ obrljiv.

$d < -1 \Rightarrow 1, -1$ sta edina obrnljiva v $\mathbb{Z}[\sqrt{d}]$

x obrnljiv $\Rightarrow N(x)\, N(y) = \pm 1$

$N(a+b\sqrt{d}) = (a+b\sqrt{d})(a-b\sqrt{d}) = a^2 - b^2 d = a^2 + b^2 |d|$
≥ 1

Če je $b^2 > 1 \Rightarrow N > 1$

$\Rightarrow b = 0$

a^2 mora biti $1 \Rightarrow a = \pm 1$ ota edini

manjši

in vemo da sta drugačna to nej te je rezultat

Pokaži da so $1+i$, $7+8i$, 3 nerazcepni v $\mathbb{Z}[i]$

Recimo da so razcepni

$1+i = x\,y \quad N(xy) = x\sigma(x)\, y\sigma(y) = 2$

prasta

$p \to$ nerazcepne

$N(7+8i) = 49 + 64 =$ velik $= 113$ prasta

~~Ujeprasta~~ Nemu je 9

$3 = xy \quad N(xy) = x\sigma(x)\, y\sigma(y) = 9$

vsaj 2 morata biti 1

$\cancel{\sigma(x)} N(x) = N(\sigma(x)) = \sigma(x)\sigma\sigma(x) = \cancel{\sigma} \; x\sigma(x)$

$N(x) = n \Rightarrow n$ razcepen

$N(x) = n = x\sigma(x)$

6) Poišči vse delitelje elementa 2 v $\mathbb{Z}[i]$

$x \mid 2 \Leftrightarrow 2 = xy$ za nek y

$N(x \cdot y) = 4 \qquad N(x)N(y) = 4$

$x\sigma(x)\, y\sigma(y) = 4 = 2 \cdot 2 = xy\sigma(xy) = 2\sigma(xy)$

$\Rightarrow \sigma(xy) = 2 = \sigma(x)\sigma(y) \Rightarrow$

$N(x) = 2 \qquad x = \pm 1 \pm i$

$x = \pm 2$

$x = \pm 2i$

$\Rightarrow$ vse možnosti

1)

$a = 3+4i$ $\qquad \delta = m^2+n^2$

$b = 1-3i$

$|a| = 9+16 = 25$

$|b| = 1+9 = 10$ $\qquad$ superi della b

$a = kb + r$

$\frac{a}{b} = k + \frac{r}{b}$

$\in \mathbb{Z}[i]$ $\qquad \in \mathbb{Q}[i]$

$a = 3-4i$

$b = 1-3i$

$$\frac{a}{b} = \frac{(3-4i)(1+3i)}{10} = \frac{3+12+i(9-4)}{10} = 1 + \frac{1}{2} + \frac{1}{2}i$$

$$\frac{1}{2}(1+i)(1-3i) = \frac{1}{2}(1+3+i(1-3)) = 2-i$$

$a = b + 2-i$ $\qquad / : (2-i)$

$$\frac{a}{2-i} = \frac{b}{2-i} + 1 \quad \Rightarrow$$

$$\frac{a}{2-i} - \frac{b}{2-i} = 1 \quad \Rightarrow \quad a-b = 2-i$$

2)

$\mathbb{Z}[\sqrt{-5}]$

$a=6$

$b=2+2\sqrt{-5}$

$c=2 \Rightarrow N(a)=\frac{36}{2}=3 \quad N(b)=\frac{4+4\cdot 5}{2}=4\cdot 6$

$=1+\sqrt{5}$

$\varepsilon|c \Rightarrow N(\varepsilon)|N(c)=5 \Rightarrow \varepsilon \in \{\pm 1, \pm i\}$, $d|\frac{b}{d}=\frac{a}{2}$, $d|b \Rightarrow 6\cdot 2\cdot 3 \mid 6\cdot 2\cdot 3$

$\partial(c)=1+5=6$

$6^2 \qquad 6\cdot 4$

$N(\gcd(a,b))=\gcd(N(a),N(b))=\gcd(36,24)=12=x^3\cdot 5y^2$

$y^2\in\{1,2\}$

$\Downarrow \qquad \Downarrow$

$x^2=2\cdot 3 \times \qquad x^2=x$

Take nedostoje

3) UFD $\Rightarrow$ gcd dobrze

$a = \prod_{i \in I} p_i^{k_i}$ $\quad d = \gcd(a,b) = \prod_{i \in I} p_i^{\min\{k_i, l_i\}}$

$b = \prod_{i \in I} p_i^{l_i}$

$d | a \wedge d | b$

Niech bo $c = \prod_{i \in I} p_i^{m_i}$ $\quad c | a \wedge c | b$

$a = c\alpha \quad \alpha = \prod p_i^{n_i}$

$a = \prod p_i^{n_i + m_i} = \prod p_i^{k_i} \Rightarrow m_i \leq k_i$

podobnie $m_i < l_i \Rightarrow \forall_i$

$c | d$

$\mathbb{Z}[\sqrt{-2}]$ euklidisch

a) $\forall a,b \; b \neq 0 \; \exists q,r \in R.\; a = qb + r \wedge r = 0 \vee \delta(r) < \delta(b)$

b) $\delta(a) \leq \delta(a,b) \; \forall a, b \in R - \{0\}$

$\delta: m + \sqrt{-2}n = m^2 + 2n^2$

Venn der b vefa

$a, b \in \mathbb{Z}[\sqrt{-2}]$ [illegible] $a = x + \sqrt{-2}y \quad b = m + \sqrt{-2}n$

$$\frac{a}{b} = q_1 + q_2\sqrt{-2}$$

$$a = \left([q_1] + [q_2]\sqrt{-2}\right)b + \left(q_1 - [q_1] + (q_2 - [q_2])\sqrt{-2}\right)b$$

DN:

Niech bo $K[X]$ UFD

Potem ma pdinem n^{ek} endiczo tolbarzero

Zaraz je tud $K[X]$ UFD

Niech bo K UFD

Niech bo $p(x)$ pdjuken pdinom

Razcepim pots $= \prod_{k=1}^{r}\left(\sum_{i=1}^{n} a_{ik}x^{n}\right) =$

$= \prod_{k=1}^{r}\left(a_{nk}x^{n} + \dots + a_{0k}\right) = x^{nr}\left(\underbrace{\prod_{k=1}^{r} a_{nk}}\right) + \dots$

$\mathrm{cont}(fg) = \mathrm{cont}(f)\,\mathrm{cont}(g)$

$p \mid \mathrm{cont}(f,g) \Rightarrow \gcd(a_0, \dots, a_n) = p$

$fg = 0 \vee K/p\,[X]$

$\Rightarrow f = 0 \vee g = 0$

$(ax+b)(cx+d) = acx^2 + x(cb+ad) + bd$

$\Rightarrow p \mid \mathrm{cont}\, f \vee p \mid \mathrm{cont}\, g$

Remarks f rozkłada FOX

płaszczyzna zapisana $af = p_1 \dots p_n \in K[x]$

Najpierw dzielę $\mathrm{cont}(af)$ $\ker af = pq$ / cont

$$\mathrm{cont}(af) = \mathrm{cont}(g)\,\mathrm{cont}(p) = a\,\mathrm{cont}\,f$$

$= a \cdot$ end: Oto

end: Oto chyba dobrzeje

$$a \mid \mathrm{cont}(pq) \Rightarrow \frac{p}{x} \cdot \frac{pq}{y} = \frac{fa}{a} = f$$

$\underbrace{x}_{\vee p} \mid \underbrace{y}_{\vee q}$

$f \mid pq \Rightarrow f \mid p \vee f \mid q$

$af = pq$ $\quad \mathrm{cont}\,f\,\mathrm{cont}\,k = \mathrm{cont}\,p\,\mathrm{cont}\,q$

pq irred. rozkład $\Rightarrow f \mid p \vee f \mid q$

$\Rightarrow \forall p_1, \dots, p_n$

2) $K[X]$ glavni ideal $\Leftrightarrow$ K je polje

$a \in K - \{0\}$

$\exists b$ $\underline{ab = 1}$

(a) je ideal $(a) = \{af : f \in K[X]\}$

20.10

1) $1+\sqrt{-3}$ nerazcepen v $\mathbb{Z}[\sqrt{-3}]$,
ampak ni praelement

$1+\sqrt{3}$ nerazcepen

Recimo da je $(1+\sqrt{-3}) = \alpha \cdot \beta$

$N(\alpha \cdot \beta) = N(\alpha) \cdot N(\beta) = 1+3 = 4$

$N(\alpha) = N(\beta) = \pm 2$
ĉe bi bila 1 bi bila obrnljiva

$N(a+\sqrt{-3}b) = a^2 + 3b^2 = \pm 2$

$b = 0$
$a^2 = \pm 2$ ne obstaja

Ni praelement

$\cancel{(a+b\sqrt{-3})}(1+\sqrt{-3}) = \cancel{a+b\sqrt{-3}+a\sqrt{-3}-b} =$

Vsak element deli svojo normo

$(1+\sqrt{-3})(1-\sqrt{-3}) = 4 = 2 \cdot 2$

$(1+\sqrt{-3}) \nmid 2$

Recimo da $k(1+\sqrt{-3}) = 2$ $\quad N(2) = 4 \Rightarrow$
$N(k) = \pm 1$
$\Rightarrow k = \pm 1$

Domača naloga

1)

$$\omega = \frac{(1+\sqrt{-19})}{2} \qquad \mathbb{Z}[\omega] = \{a+b\omega : a,b \in \mathbb{Z}\}$$

$$N(a+b\omega) = |a+b\omega|^2 = a^2+ab+5b^2$$

2 in 3 nerazcepna

$N(2) = 4$

$2 = ab$; a,b neobr. $\Rightarrow N(a) = N(b) = 2$

$a = a+b\omega$

$N(a) = a^2+ab+5b^2 = 2$

$b \neq 0 \Rightarrow 5b^2 > 2 \Rightarrow N(a) > 2$ ✗

$N(3) = 9 \qquad 9 = x \cdot y \qquad 9 = 3 \cdot 3 \Rightarrow N(x) = N(y) = 3$

$N(x) = 3 \qquad x = a+b\omega$

$a^2+ab+5b^2 = 3$

$b \neq 0 \Rightarrow 5b^2 > 3 \Rightarrow N(x) > 3$ ✗

$N(x) = 1 \Rightarrow x \in \{1,-1\}$

$a^2+ab+5b^2 = 1$

$b \neq 0 \Rightarrow 5b^2 > 1$ ✗ $\Rightarrow b = 0$

$a^2 = 1 \Rightarrow a \in \{1,-1\} \Rightarrow x \in \{1,-1\}$

Moduli

$K = \{$zgornje trikotne matrike nad $F\}$

$M = \begin{bmatrix} F \\ F \end{bmatrix}$ je levi K-modul

iščemo podmodule od M

$KM \subseteq M$

$N = \{\lambda \begin{bmatrix} u \\ v \end{bmatrix} ; \lambda \in F\}$ za $u, v \in F$

$$\begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \cdot \lambda \begin{bmatrix} u \\ v \end{bmatrix} = \lambda \begin{bmatrix} au+bv \\ cv \end{bmatrix} \in N$$

$$\lambda au + \lambda bv = \mu u$$
$$\lambda cv = \mu v \Rightarrow \lambda c = \mu$$

$$\lambda au + \lambda bv = \lambda cu$$
$$au - cu + bv = 0$$
$$bv = (c-a)u$$

DN najprej

$\mathbb{Z}[\omega]$ ni evklidski

$(\exists \sigma : \mathbb{Z}[\omega] - \{0\} \to \mathbb{N} \cup \{0\})$

$\omega = qx + r \Rightarrow r \in \{1, -1, 0\}$

$r = 0 \Rightarrow \omega = qx \Rightarrow x$ asociiran z ω

$N(x) = N(\omega) = 5$

$2 = q'x \pm 1 \Rightarrow x \in \{\pm 2, \pm 3\} \Rightarrow N(x) \in \{4, 9\}$

$r = 1 \Rightarrow \omega = qx + 1$ $\times$

$-1 + \omega = qx \Rightarrow 1 - 1 + 5 = N(q) N(x) = 5$

$N(x) = 5$ $\times$

$r = 1 \Rightarrow 1 + \omega = qx \Rightarrow 7 = N(q) N(x) \Rightarrow N(x) = 7$ $\times$

$\mathbb{Z}[\omega]$ je glavni:

$$(\forall \alpha \in \mathbb{C}-\mathbb{Z}[\omega]. \exists p, q \in \mathbb{Z}[\omega]. \ 0<|p\alpha - q|<1)$$
$$\Longrightarrow \mathbb{Z}[\omega] \text{ glavni kolobar}$$

Naj bo I poljuben ideal. Naj bo $b \in I$ tak da

$\forall x \in I. \ N(x) \geq N(b)$

Naj bo $a \in I$ poljuben $\quad \alpha = \frac{a}{b} \quad a = \alpha b$

$\alpha \in \mathbb{Z}[\omega] \Rightarrow b \mid a$ za $\forall a \in I \Rightarrow I = (b)$

$\alpha \in \mathbb{Z}[\omega]$ in $\{p, q\}$ $\Longrightarrow \quad 0<|p\frac{a}{b} - q|^2<1 \quad / \cdot |b|$

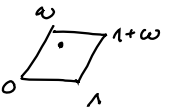
$$0<|\underbrace{pa}_{\in I} - \underbrace{qb}_{\in I}|^2<|b|^2 \quad \times$$

$\forall \alpha \in \mathbb{C}-\mathbb{Z}[\omega]\ \exists p, q \in \mathbb{Z}[\omega].\ 0<|p\alpha-q|<1$

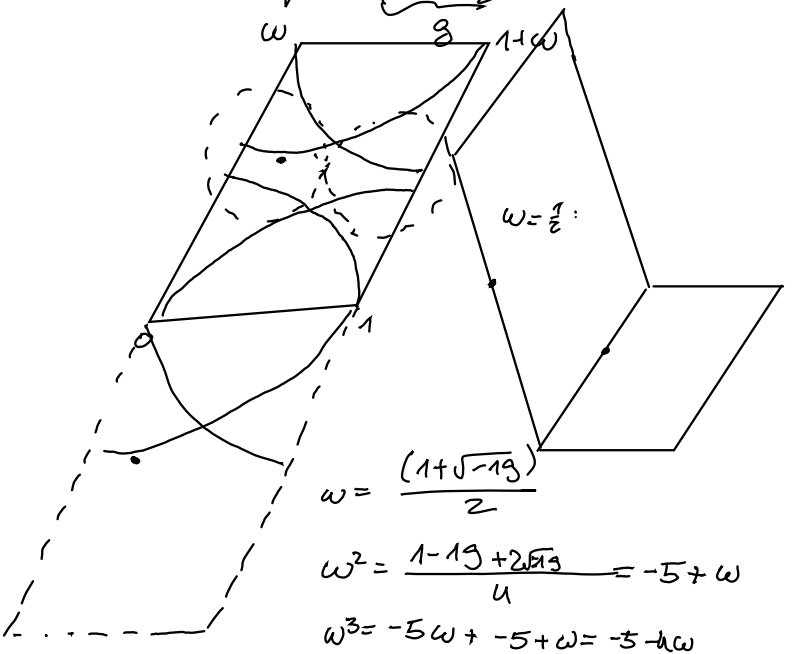
$\alpha \in \mathbb{C}-\mathbb{Z}[\omega]$ poljuban

$\alpha = a+b\omega \quad a,b\in\mathbb{R}$

$\beta = \alpha-(m+\omega n) \quad \beta \in$



$p\alpha - q = p\beta + \underbrace{\beta(\dots)-q}$



$$\omega = \frac{(1+\sqrt{-19})}{2}$$

$$\omega^2 = \frac{1-19+2\omega\sqrt{-19}}{4} = -5+\omega$$

$$\omega^3 = -5\omega + -5+\omega = -5-4\omega$$

DM za tedne:

$d = 4k+1$

$(a+b\sqrt{d})(a-b\sqrt{d}) = a^2 - db^2 = 4(2-k)$

$b^2 = 1$:

$a^2 - 4k - 1$

$a = 3$

$8 - 4k = 4(2-k)$

$b^2 = 1$

$a^2 = 9$

$\{(x-1)p(x)\} = \{\sum a_n = 0\}$

Tiote je ideal

$$\sum_{n=0}^{k} a_n x^n$$

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 : (x-1) = a_n x^{n-1}$$

$$- \; a_n x^n + a_n x^{n-1} \qquad \cancel{+a_{n-1}}$$

$$(a_n + a_{n-1}) x^{n-1} + \dots + a_0$$

ic get

kdyby pod zbytek

and . pokp.

$$ax - a = a(x-1) \checkmark$$

Ora se la base è $\exists B \in \{f_1, \dots, f_g\}$

tale che elementi che appartengono

$g_1 f_1 + \dots + g_n f_n$

$\sqrt{f_1 + \dots + f_n}$

$\sqrt{g_1 f_1 + \dots + g_n f_n} \leq \sqrt{\max(g_1, \dots)} \sqrt{f_1 + \dots + f_n}$

$\leq h_1 f_1 + \dots + h_n f_n$

$M = {}_{\mathbb{Z}}\mathbb{Z}_{12}$

$\mathbb{Z}_2 = \{0, 6\}$ podgrupa

$\mathbb{Z}_3 = \{0, 4, 8\}$ podgrupa

$\mathbb{Z}_4 = \{0, 3, 6, 9\} \supseteq \mathbb{Z}_2$

$\mathbb{Z}_6 = \{0, 2, 4, 6, 8, 10\} \supseteq \mathbb{Z}_2, \mathbb{Z}_3$

3) $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}_{12})$, $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}_3, \mathbb{Z}_4)$, $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}_{12}, \mathbb{Z}_{12})$

$\mathbb{Z} \longrightarrow \mathbb{Z}_{12}$

$0 \longmapsto 0$ 1 nebude dobrá

$1 \longmapsto$ každá $\Rightarrow$ imamo 12 homomorfizma

$$\boxed{\begin{array}{l} K \text{ kolobar, } {}_KM \text{ modul} \Rightarrow \\ \mathrm{Hom}(K, M) = \{x \longmapsto xm : m \in M\} \cong {}_KM \end{array}}$$

$\mathbb{Z}_3 \to \mathbb{Z}_4$

$0 \longmapsto 0$

1.	2.	3.
$1 \longmapsto 1$	$1 \longmapsto 2$	$n \longmapsto 0$
$2 \longmapsto 2$	$2 \longmapsto 1$	
$3 \longmapsto 0$	$3 \longmapsto 0$	
$4 \longmapsto 1$	$4 \longmapsto 2$	

$\mathbb{Z}_{12} \longmapsto \mathbb{Z}_{12}$

$0 \longmapsto 0$

$1 \longmapsto a$

$n \longmapsto na$ mod 12

K komutativní. K environment $\Leftrightarrow$ K okruh

$\Rightarrow$ Rozmyslete si environment, a ukažte, že

a absolutní

$Ka = K \Rightarrow \exists k \in K.\ ka = 1$

$\Leftarrow$ M podmodul M

Vaje za nazaj

1 Naloga M K-modul $M \neq \{0\}$ enostaven $\Leftrightarrow M = Km$ $\forall m \in M$

$\Rightarrow$ Naj bo M enostaven. Edina podmodula sta $\{0\}$ in M

Naj bo $m \in M$ poljuben $\underline{Km}$ je podmodul

$\alpha, \beta \in K$ $\alpha k m - \beta h m = (\alpha k - \beta h) m \in Km$

$\Leftarrow$ $M = Km$ za $\forall m \in M$.

Naj bo N podmodul.

$\forall n \in N$. $Kn = M$ torej $N = M$

2. naloga

levi $_K$ modul $_KM$ enostaven $\Leftrightarrow$ $M \cong K/_K I$, kjer je $_K I \leq_K$ maksimalen levi ideal

$\Rightarrow$ Naj bo M enostaven, $a \in M$

$\varphi: K \longrightarrow M$ $\quad$ φ surjektiven po prejšnji nalogi

$k \mapsto ka$

$\ker\varphi = \{k;\ ka = 0\} = {}_K I$ maksimalen

$I \subsetneq J \subsetneq K$

$K \longrightarrow M$

$J \xrightarrow{sur} \varphi_*(J)$

$I \longrightarrow 0$

$\ker J \neq \ker\varphi$

$0 \subsetneq \varphi_*(J) \subsetneq M$ $\quad$ $\times$

$\varphi_*(J) = M \Rightarrow m = ja$ poljuben

$\Rightarrow j = ja \Rightarrow j(1-a) = 0$

$1-a \in \ker\varphi$

$\Rightarrow$

$1-a \in J$

$(1-a)+a = 1 \in J$

$\Rightarrow J = K$ $\quad$ $\times$

$\Longrightarrow J = I \vee J = K$

$\Leftarrow$ Naj bo $_KM \cong K/_K I$; I maksimalen levi ideal

$$\begin{array}{ccccc} 0 & < & N & < & M \\ \uparrow & & \uparrow & & \uparrow \\ I & < & J & < & K \end{array}$$

φ surj

$J = \varphi^*(N)$

$N \subseteq \varphi_*(J) = \varphi_*(\varphi^*(N)) \Longrightarrow \varphi_*(J) = N$

$K/I \to M$

$1+I \mapsto e \in M - \{0\}$ $\quad \Longrightarrow \quad$ $\varphi: K \longrightarrow M$

$k \mapsto ke$

$x + I = x(1+I) \mapsto xe$

Enostaven $\quad$ $I = J \vee J = K$

$$\begin{array}{ccccccc} I & < & J_2 & < & J_1 & < & K \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 & < & \varphi_*(J_2) & = & \varphi_*(J_1) & < & M \end{array} \qquad /:\varphi(J_2)$$

$$0 = \frac{\varphi_*(J_1)}{\varphi_*(J_2)} < M/\varphi(J_2)$$

??

Une nebulo
espaca

Od zadnjič:

$$\mathrm{End}(M\oplus N) \cong \begin{bmatrix} \mathrm{End}(M) & \mathrm{Hom}(N,M) \\ \mathrm{Hom}(M,N) & \mathrm{End}(N) \end{bmatrix}$$

1) dolaši kolobarje endomorfizmov

$\mathrm{End}_{\mathbb{Z}}(\mathbb{Z}_2\oplus\mathbb{Z}_3)$, $\mathrm{End}_{\mathbb{Z}}(\mathbb{Z}_6\oplus\mathbb{Z}_3)$

$$\mathrm{End}_{\mathbb{Z}}(\mathbb{Z}_2\oplus\mathbb{Z}_3) = \begin{bmatrix} \mathrm{End}\,\mathbb{Z}_2 & \mathrm{Hom}(\mathbb{Z}_2,\mathbb{Z}_3) \\ \mathrm{Hom}(\mathbb{Z}_3,\mathbb{Z}_2) & \mathrm{End}\,\mathbb{Z}_3 \end{bmatrix}$$

$\mathrm{End}\,\mathbb{Z}_2$: $0\mapsto 0$, $1\mapsto 1$ $\quad$ $1\mapsto 0$, $0\mapsto 1$ $\quad \cong \mathbb{Z}_2$

$\boxed{\text{Splošno: } \mathrm{End}\,\mathbb{Z}_p \cong \mathbb{Z}_p}$ $\quad \Rightarrow \mathrm{End}\,\mathbb{Z}_3 \cong \mathbb{Z}_3$

$1\mapsto$ red elementa 3 ne obstaja!

$$\mathrm{End}(\mathbb{Z}_2\oplus\mathbb{Z}_3) \cong \begin{bmatrix} \mathbb{Z}_2 & 0 \\ 0 & \mathbb{Z}_3 \end{bmatrix} \cong \mathbb{Z}_2\times\mathbb{Z}_3 \cong \mathbb{Z}_6$$

$\mathrm{End}(\mathbb{Z}_6\oplus\mathbb{Z}_3)$: $\mathrm{End}(\mathrm{End}(\mathbb{Z}_2\oplus\mathbb{Z}_3)\oplus\mathbb{Z}_3) =$

$$= \begin{bmatrix} \mathrm{End}(\mathbb{Z}_2\oplus\mathbb{Z}_3) & \mathrm{Hom} \\ \mathrm{Hom} & \mathrm{End}\,\mathbb{Z}_3 \end{bmatrix} = \begin{bmatrix} \mathbb{Z}_6 & \mathrm{Hom} \\ \mathrm{Hom} & \mathbb{Z}_3 \end{bmatrix}$$

$$\mathrm{Hom}(\mathbb{Z}_6,\mathbb{Z}_3) \underset{\mathbb{Z}}{=} \mathrm{Hom}(\mathbb{Z}_2\oplus\mathbb{Z}_3,\mathbb{Z}_3) \cong$$

$$\cong \underbrace{\mathrm{Hom}(\mathbb{Z}_2,\mathbb{Z}_3)}_{0}\oplus\mathrm{Hom}(\mathbb{Z}_3,\mathbb{Z}_3) \cong \mathbb{Z}_3$$

$$\mathrm{End}(\mathbb{Z}_6\oplus\mathbb{Z}_3) = \begin{bmatrix} \mathbb{Z}_6 & \mathbb{Z}_3 \\ \mathbb{Z}_3 & \mathbb{Z}_3 \end{bmatrix}$$

$1\mapsto 2$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} \overset{?}{=} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \text{ se ne izide}$$

$x \mapsto (k, x \bmod \mathbb{Z}_3)$

$\mathbb{Z}_3 \to \mathbb{Z}_2\oplus\mathbb{Z}_3 \to \mathbb{Z}_6$

$$\begin{bmatrix} 0 & \\ 0 & 0 \end{bmatrix} \quad 1\mapsto(1,0)\mapsto 4 \qquad \begin{bmatrix} 0 & 0 \\ \mathbb{Z}_6 \to \mathbb{Z}_2+\mathbb{Z}_3 \to \mathbb{Z}_3 & 0 \end{bmatrix}$$

$(1,0)\mapsto 2$

$x \mapsto 2x \bmod 3$

$$= \begin{bmatrix} x\mapsto 2x\mapsto(2x,0) \overset{\bmod 3}{=} 4x \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

$\updownarrow \mathbb{Z}_6 \to \mathbb{Z}_3$ $\qquad \updownarrow$

$$\begin{bmatrix} 0 & x\mapsto 4x \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 \\ x\mapsto 2x & 0 \end{bmatrix} \qquad \begin{bmatrix} x\mapsto 8x & 0 \\ 0 & 0 \end{bmatrix}$$

$\mathbb{Z}_3 \to \mathbb{Z}_6$

k: $\mathbb{Z}_3 \to \mathbb{Z}_6$ $\qquad$ $\mathbb{Z}_6 \to \mathbb{Z}_3$

$1 :=$ $1\mapsto(1,0)\to 4$ $\qquad$ $k = 1\mapsto k \bmod 3$

$2 :=$ $1\mapsto(2,0)\mapsto 2$

$k :=$ $1\mapsto(k,0)\mapsto \ldots$

Hitreje:

$$\mathbb{Z}_{18} = \mathbb{Z}_6 \oplus \mathbb{Z}_3 = \mathbb{Z}_2 \oplus (\mathbb{Z}_3 \oplus \mathbb{Z}_3)$$

$$\mathrm{End}(\mathbb{Z}_6 \oplus \mathbb{Z}_3) = \begin{bmatrix} \mathbb{Z}_2 & \mathrm{Hom}(\mathbb{Z}_3 \oplus \mathbb{Z}_3, \mathbb{Z}_2) \\ \mathrm{Hom}(\mathbb{Z}_2, \mathbb{Z}_3 \oplus \mathbb{Z}_3) & \mathrm{End}(\mathbb{Z}_3 \oplus \mathbb{Z}_3) \end{bmatrix} =$$

$$\mathrm{Hom}(\mathbb{Z}_2, \mathbb{Z}_3) \oplus \mathrm{Hom}(\mathbb{Z}_2, \mathbb{Z}_3)$$

$$= \begin{bmatrix} \mathbb{Z}_2 & 0 \\ 0 & \begin{bmatrix} \mathbb{Z}_3 & \mathbb{Z}_3 \\ \mathbb{Z}_3 & \mathbb{Z}_3 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} \mathbb{Z}_2 & 0 & 0 \\ 0 & \mathbb{Z}_3 & \mathbb{Z}_3 \\ 0 & \mathbb{Z}_3 & \mathbb{Z}_3 \end{bmatrix} \cong$$

$$\cong \mathbb{Z}_2 \oplus M_2(\mathbb{Z}_3)$$

Bimoduli

$K^{op} = \{ x^{op} \mid x \in K \}$ $\qquad x^{op} \cdot y^{op} = (yx)^{op}$

Grupe: $G \cong G^{op}$
$x^{op} = x^{-1}$

${}_KM \cong M_{K^{op}}$ vsak levi modul lahko predstavimo kot desni K modul

Bimodul je ${}_KM_S$, povezuje pa tudi asociativnostni zakon
K, S kolobarja $\quad {}_KM$ je levi K-modul
M_S je desni S-modul

$x \in K, \; y \in S$

$m \in M \quad \underbrace{(x \cdot m)}_{\in M} y = x(m \cdot y)$

Primeri:

1) ${}_KK_K$ je bimodul

2) ${}_KM_{K^{op}}$ ni bimodul (razen če K komut.

$m \cdot x^{op} := xm \qquad x(my^{op}) = xym$
$\qquad\qquad\qquad\qquad \parallel$
$\qquad\qquad\qquad\qquad (xm)y^{op} = yxm$

3) ${}_KM_{End(M)^{op}}$ je bimodul

$(xm)\varphi^{op} = \varphi(xm) = x\varphi(m) = x(m\varphi^{op})$

2) Naj bosta ${}_kM_S$ in ${}_kN_R$ bimodula (k, S, R kolobarji)

Premisli, da imajo homomorfizmi $M \to N$ poleg ene strukture (S, R)-bimodula

homomorfizmi kot levih k-modula

zgubi strukturo k modula

${}_S\mathrm{Hom}(M,N)_R$

$s \in S \quad s\varphi := m \mapsto \varphi(ms)$

$r \in R \quad \varphi r := m \mapsto \varphi(m)r$

Hom(M,N) je levi S modul

$$(s+t)\varphi = (m \mapsto \varphi(m(s+t))) = (m \mapsto \varphi(ms) + m \mapsto \varphi(mt))$$

$$\varphi(m(s+t)) = \varphi(ms) + \varphi(mt) \qquad = s\varphi + t\varphi$$

$$1 \cdot \varphi = (m \mapsto \varphi(m)) = \varphi$$

$$s(\varphi + \psi) = \checkmark$$

$\forall m.$ $(st)\varphi(m) = \varphi(mst) = \varphi((ms)t) =$

$= (t\varphi)(ms) = s(t\varphi(m)) = s(t\varphi)(m)$

ali je to vredu $(t\varphi)(xm) = x((t\varphi)(m))$

$$(t\varphi)(xm) = \varphi((xm)t) = \varphi(x(mt)) = x\varphi(mt)$$

Podobno za desni R modul

$$s(\varphi r)(m) = (\varphi r)(ms) = \varphi(ms)r = (s\varphi(m))r$$

Lev K-modul ${}_K M$. Duál je $M^*_K = \mathrm{Hom}({}_K M, {}_K K_K)$

Funkcional

Ker je domena bimodul, je tudi M^*_K desni K modul

$$M = \begin{bmatrix} F \\ F \end{bmatrix}$$

3) Dobiši $\left({}_{M_2(F)} M\right)^*$

$$M^*_{M_2(F)} = \mathrm{Hom}\left({}_{M_2(F)} M, {}_{M_2(F)} M_2(F)_{M_2(F)}\right) \cong$$

$$\begin{bmatrix} F & F \\ F & F \end{bmatrix} \cong \begin{bmatrix} F & 0 \\ F & 0 \end{bmatrix} \oplus \begin{bmatrix} 0 & F \\ 0 & F \end{bmatrix} \cong M \oplus M$$

$$\cong \mathrm{Hom}(M, M \oplus M) = \mathrm{End}(M) \oplus \mathrm{End}(M) \cong F^2$$

To sta ob seaq (shurova lema)

$M=\begin{bmatrix} F \\ F \end{bmatrix}$ $\qquad M \overset{?}{=} [F\ F]$

$K=\begin{bmatrix} F & F \\ F & F \end{bmatrix}$ $\qquad K \cong M \oplus M$ vrstica + vrstica

$$\begin{bmatrix} 0 & F \\ 0 & F \end{bmatrix} \oplus \begin{bmatrix} F & 0 \\ F & 0 \end{bmatrix}$$

struktura desnega K-modula

$$M^* = \mathrm{Hom}(M,K)_K \cong \mathrm{Hom}(M,M) \oplus \mathrm{Hom}(M,M) \cong F \oplus F$$

ni desni K modul

izgubimo strukturo modula

(*) $F \cong \mathrm{Hom}(M,M) = \mathrm{End}\,M$

$\mathrm{End}(M)$ obseg, če M enostaven

$F \cong \mathrm{End}(M)$

$\supseteq$ V posebnem: $\varphi \in \mathrm{Hom}_K(M,M) \leq \mathrm{Hom}_F(M,M) \cong M_2(F)$ 2-dim vek. prostor

$\varphi(Ax) = A\varphi(x) \qquad \varphi = B \Rightarrow$

$BAx = ABx$ za $\forall A \in K$ $\forall x \in M$. $\Rightarrow BA = AB \Leftrightarrow \forall A \in K$.

$B \in Z(M_2(F)) = F \cdot I \cong F$ diagonalne matrike

$\subseteq$ očitno

(**) M^*_K tvoj $(F \oplus F)_K$

$\varphi \in \mathrm{Hom}(M,K)$

$\varphi(m) = (\varphi_1(m), \varphi_2(m)) = \left(\varphi(m)\cdot\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \varphi(m)\cdot\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$

$\leftarrow$ stolpca

$\varphi(m) = \left[\varphi_1(m) \,\vdots\, \varphi_2(m)\right]$

$\varphi(Am) = \left[\varphi_1(Am) \,\vdots\, \varphi_2(Am)\right]$

$A\varphi(m) = \left[A\varphi_1(m) \,\vdots\, A\varphi_2(m)\right]$ $\qquad \varphi_{1,2}$ skalarna vektorja identitete

$A = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \qquad \varphi \cdot A = (m \longmapsto \varphi(m)\cdot A)$ $\in M^*$ $\qquad$ dano po definiciji

$[\varphi_1, \varphi_2]\cdot A = \varphi_1 \cdot a + \varphi_2 \cdot b + \varphi_1 \cdot c + \varphi_2 \cdot d = \varphi_1(a+c) + \varphi_2(b+d)$

kako sklepamo da je

$A = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \qquad \varphi(m) = [\varphi_1(m), \varphi_2(m)]$ * $\quad m = \begin{bmatrix} x \\ y \end{bmatrix}$

skalarja

$= [\lambda_1 m \quad \lambda_2 m] = \begin{bmatrix} \lambda_1 x & \lambda_2 x \\ \lambda_1 y & \lambda_2 y \end{bmatrix}$

$$\varphi(m)\cdot A = \begin{bmatrix} a\lambda_1 x + b\lambda_2 x & c\lambda_1 x + d\lambda_2 x \\ a\lambda_1 x + b\lambda_2 y & c\lambda_1 y + d\lambda_2 y \end{bmatrix} =$$

$$= \begin{bmatrix} x(a\lambda_1 + b\lambda_2), & x(c\lambda_1 + d\lambda_2) \\ y(a\lambda_1 + b\lambda_2) & y(c\lambda_1 + d\lambda_2) \end{bmatrix}$$

$\varphi = [\lambda_1 \quad \lambda_2] \qquad \varphi \cdot A = [\lambda_1 a + \lambda_2 b, \quad c\lambda_1 + d\lambda_2]$

Lažje: $\quad F \oplus F \longrightarrow \mathrm{Hom}(M,K)$

$$[\lambda, \mu] \longmapsto \left(\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x \\ y \end{bmatrix}[\lambda, \mu]\right)$$

$$(\varphi \cdot A)\begin{bmatrix} x \\ y \end{bmatrix} = \Big(\underbrace{\begin{bmatrix} x \\ y \end{bmatrix}[\lambda, \mu]}_{\varphi}\Big)A = \begin{bmatrix} x \\ y \end{bmatrix}\big([\lambda, \mu]A\big)$$

1) Dokaži dual ${}_{\mathbb{Z}}\mathbb{Z}_n^*$

$= \mathrm{Hom}(\mathbb{Z}_n, \mathbb{Z}) = 0$

$1 \longmapsto x \qquad \mathrm{red}\,1 \mid \mathrm{red}\,x \Rightarrow x = 0$

2) K cel kolobar (brez del. nič., komut)

za M K-modul definiramo

$$\mathrm{tor}(M) = \{m \in M;\ xm = 0,\ x \in K - \{0\}\}$$

$$\mathrm{tor}\left(\bigoplus_{n=2}^{\infty} \mathbb{Z}_n\right) = \bigoplus_{n=2}^{\infty} \mathbb{Z}_n$$; ampak ne obstaja element, ki bi uničil vse

Dokaži da je $\mathrm{tor}(M) \leq M$ podmodul in

$\varphi: M \to N$

$\varphi_*(\mathrm{tor}(M)) \leq \mathrm{tor}(N)$

a) $xm = 0 \quad yn = 0 \quad m, n \in M \quad x, y \in K$

$\underbrace{xy}_{\neq 0}(m+n) = xym + x\underbrace{yn}_{0} = yxm = 0$

$\alpha \in K \quad xm = 0$

$x\,\alpha m = \alpha x m = 0$

b) φ homomorfizem

$x\varphi(m) = \varphi(xm) = 0 \to m \in \mathrm{tor}(M) \Rightarrow \varphi(m) \in \mathrm{tor}(N)$

$\Rightarrow \varphi_*(\mathrm{tor}(M)) \subseteq \mathrm{tor}(N)$

je podmodul: $xy(\alpha\varphi(m) + \beta\varphi(n)) =$

$\alpha y \underbrace{\varphi(xm)}_{0} + \beta x \underbrace{\varphi(n)}_{0} = 0$

$\mathrm{tor}\left(M/\mathrm{tor}(M)\right) = 0$ DN

$$x(m + \mathrm{tor}M) = \underbrace{xm}_{} + \mathrm{tor}M$$

$= 0 \Rightarrow m \in \mathrm{tor}M \rightarrow m + \mathrm{tor}M = 0$

$\neq 0 \Rightarrow m \notin \mathrm{tor}M$

3. M je prost $\overset{K \text{ cel}}{\Downarrow} \Rightarrow \operatorname{tor}(M)=0$

$$\varphi : M \xrightarrow{\cong} \bigoplus_{i=I} K$$

$$\varphi(\operatorname{tor} M) \leq \operatorname{tor}\left(\bigoplus_{i=I} K\right)$$

φ inj $\Rightarrow$ dovolj je dokazat $\operatorname{tor}\left(\bigoplus_{i=I} K\right)=0$

(Dovolj je inj vložitev, ne izomorfizem)

$$\bigoplus_{i\in I} K = \{(x_i)_{i\in I} \,;\, x_i \in K \quad x_i = 0 \text{ za vse razen konono mnogo}\}$$

$$(x_1, x_2, \ldots \ldots x_i, \ldots)$$

$$(x_i)_{i\in I} \in \operatorname{tor}\left(\bigoplus K\right)$$

$$y(x_i)_{i\in I} = (yx_i)_{i\in I} \Leftrightarrow \overset{0}{\overset{\shortparallel}{yx_i}} = 0 \ \forall i$$

ne vel, a za noben element razen 0

$$\Rightarrow x_i = 0 \ \forall i$$

4) Al: je $\mathbb{Z}_n$ prost $\mathbb{Z}$-modul?

NE, ker ima torzijo

$$\boxed{\begin{array}{l}\text{Izrek: } M \text{ končno generiran } \mathbb{Z}\text{-modul.} \\ \qquad M \text{ prost} \Leftrightarrow M \text{ torzijsko prost}\end{array}}$$

5) Pokaži

${}_{\mathbb{Z}}\mathbb{Q}$ ni prost $\mathbb{Z}$-modul

$\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$

$(cb)\frac{a}{b} - (ad)\frac{c}{d} = 0$ netrivialna linearna kombinacija

Al: je lahko baza?

$\mathbb{Q} = \left(\frac{c}{b}\right)$ $\quad \frac{1}{b^2}$ ni oblike $\alpha \frac{a}{b}$

Al: je $(\mathbb{Q}^*, \cdot)$ prost $\mathbb{Z}$ modul

$-1 \in \mathbb{Q}^*$ $\quad (-1)(-1) = 1 \Rightarrow (-1) \in T(\mathbb{Q}^*)$

$\Rightarrow$ ima torzijo, torej ni prost

Kaj pa $(\mathbb{Q}_+^*, \cdot)$

$\sum \left(\frac{a_i}{b_i}\right)^{n_i} > 0 \neq 0$

Eksaktna zaporedja

kratka ≡ kvocienti

$$0 \xrightarrow{0} L \xrightarrow{\varphi} M \xrightarrow{\psi} N \xrightarrow{0} 0$$

$\operatorname{im}\varphi = \ker\psi \qquad \operatorname{im}\psi = \ker 0 = N$

$0 = \operatorname{im} 0 = \ker\varphi$

ψ surjektivna

φ injektivna

$$M/\varphi(L) \cong N$$

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}_2 \longrightarrow 0$$

$x \longmapsto 2x$

$-2x \qquad x \longmapsto x \bmod 2$

inj

edina možna preslikava

$\ker\psi = 2\mathbb{Z} = \operatorname{im}\varphi$

enotna krožnica

$$0 \xrightarrow{\text{inj}\,\checkmark} \mathbb{Z} \longrightarrow \mathbb{R} \longrightarrow \mathbb{T} \xrightarrow{\text{sur}\,\checkmark} 0$$

$n \longmapsto \boxed{\begin{matrix} n \\ x \end{matrix}} \begin{matrix} \longmapsto 1 \\ \longmapsto e^{2\pi i x} \end{matrix}$

P je projektiven $\iff$ $\forall$ eksaktno zaporedje $0 \to L \to M \to P \to 0$ razpade

kratko eksaktno zaporedje razpade $\iff$ je ekvivalentno $0 \to L \to L \oplus N \to N \to 0$

P je projektiven $\iff$ 1) $\iff$ 2)

$\iff$ 1)

$$\begin{tikzcd} & P \arrow[dl, dashed, "\exists g"'] \arrow[d, "\forall f"] & \\ A \arrow[r, "sur"'] & B \arrow[r] & 0 \end{tikzcd}$$

$\iff$ 2) P je direktni sumand prostega modula $K^{(I)}$

$$\exists N.\ P \oplus N \cong K^{(I)} = \bigoplus_{i \in I} K$$

kolobar

1. naloga 1.12

$\begin{bmatrix} F \\ F \end{bmatrix}$ je projektiven in ne prost nad $M_2(F)$

2) $M_2(F) \cong \begin{bmatrix} F \\ F \end{bmatrix} \oplus \begin{bmatrix} F \\ F \end{bmatrix} \cong \begin{bmatrix} F & 0 \\ F & 0 \end{bmatrix} \oplus \begin{bmatrix} 0 & F \\ 0 & F \end{bmatrix}$

zunanji $\leftarrow$ notranji

$$\begin{bmatrix} x & z \\ y & w \end{bmatrix} \longmapsto \left(\begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} z \\ w \end{bmatrix} \right) \longmapsto \begin{bmatrix} x & 0 \\ y & 0 \end{bmatrix} \oplus \begin{bmatrix} 0 & z \\ 0 & w \end{bmatrix}$$

je očiten izomorfizem

$\Rightarrow$ je projektiven

Recimo da je prost, potem

$$\begin{bmatrix} F \\ F \end{bmatrix} = \begin{cases} M_2(F) = \dim 4 \\ M_2(F) \oplus M_2(F) = \dim 8 \\ \vdots \end{cases}$$

vse dimenzije so pravilne

$\Rightarrow \begin{bmatrix} F \\ F \end{bmatrix}$ ni prost

2. način:

P proj $\Rightarrow P$ torz prost

$\varphi \quad P \longrightarrow M$ homomorfizem

$\varphi(\mathrm{tor}(P)) \subseteq \mathrm{tor}(M)$

2) P proj. $\underset{(K \text{ cel})}{\Longrightarrow}$ P torz. prost

Velja: $\varphi : P \to M$ homo.

$\varphi(\mathrm{tor}(P)) \subseteq \mathrm{tor}(M)$

$P \oplus N \cong K^{(I)}$

$P \longrightarrow K^{(I)}$

$\varphi(\mathrm{tor}) \subseteq \mathrm{tor}(K^{(I)}) = 0$ ker je $K^{(I)}$ prost

Zaradi injektivnosti φ je tudi $\mathrm{tor}(P) = 0$

$$\boxed{\begin{tikzcd} \text{prost} \arrow[r, Rightarrow] & \text{proj.} \arrow[r, Rightarrow, "K \text{ cel}"] & \text{torz. prost} \arrow[ll, Rightarrow, bend left, "K \text{ glavni: } V"] \end{tikzcd}}$$

konačnogeneriran modul

Kategorije

Top, Grp, Ab, $Vect_F$, K-Mod, ...

K mod:

objekti: K-moduli

morfizmi: homomorfizmi K-modulov

$$\mathrm{Hom}({}_K P, _) : K\ mod \longrightarrow Ab$$
$$ {}_K M \longmapsto \mathrm{Hom}(P, M)$$
$$(M \xrightarrow{f} N) \longmapsto \mathrm{Hom}(P, M) \longrightarrow \mathrm{Hom}(P, N)$$
$$\varphi \longmapsto f \circ \varphi$$

3. naloga

$\mathrm{Hom}(P, _)$ je funktor

- ohranja kompozitum
- ohranja identiteto

$$A \xrightarrow{f} B \xrightarrow{g} C$$

$$F(g \circ f) \underset{(X)}{=} (\varphi \mapsto \varphi \circ g \circ f)$$

$$F(g) \circ F(f)(X) = (\varphi \mapsto \varphi \circ g)(\varphi \mapsto \varphi \circ f)(X) =$$

$$= (\varphi \mapsto \varphi \circ g)\ X \circ f = X \circ f \circ g = (\varphi \mapsto \varphi \circ (f \circ g)) = F(f \circ g)$$

dps, zamenjan vrstni red

$$F(id) = (\varphi \mapsto id \circ \varphi) = id$$

P proj $\iff$ $\underbrace{\mathrm{Hom}(P,-)}$ eksakten

$$0 \to L \xrightarrow{\varphi} M \xrightarrow{\psi} N \to 0$$

$$0 \to \mathrm{Hom}(P,L) \xrightarrow{F(\varphi)} \mathrm{Hom}(P,M) \xrightarrow{F(\psi)} \mathrm{Hom}(P,N) \to 0$$

$\mathrm{Hom}(P,-)$ je eksakten če slika eksaktna zaporedja v eksaktna

$\Longrightarrow$ 1) $P = k$ $\quad \mathrm{Hom}(k, \underline{M}) \cong {}_kM$

$$p \mapsto p \circ \varphi \qquad \varphi \mapsto \varphi(1)$$

$$\begin{tikzcd}
\mathrm{Hom}(P,L) \arrow[r, "F(\varphi)"] \arrow[d, equal, "\cong"] & \mathrm{Hom}(P,M) \arrow[d, equal, "\cong"] & p \arrow[d, mapsto] \\
L \arrow[r, "\varphi"'] & M & \varphi(1)
\end{tikzcd}$$

2) P je prost $\quad P \cong k^{(I)}$

$$(\mathrm{Hom}, -) = \mathrm{Hom}(\bigoplus_I k, -) = \bigoplus_I \mathrm{Hom}(k, -)$$

$$0 \to \bigoplus \mathrm{Hom}(k,L) \to \bigoplus \mathrm{Hom}(k,M) \to \dots$$

$$(l_1 \dots l_n) \mapsto (\varphi(l_1), \varphi(l_2) \dots)$$

8. 12

1. ura

Tenzorski produkt

n: nujno, ampak je bolj enostavno da je

${}_KM$, ${}_KN$ modula, K komutativen

$$M \otimes_K N = \left\{ \sum_{i=1}^{l} m_i \otimes n_i \;\middle|\; m_i \in M \;\; n_i \in N \;\; l \in \mathbb{N}_0 \right\}$$

1) bilinearnost

$$(m+m') \otimes n = m \otimes n + m' \otimes n$$

$$m \otimes (n+n') = m \otimes n + m \otimes n'$$

2) biaditivnost $x \in K$

$$x \sum m_i \otimes n_i = \sum x(m_i \otimes n_i) = \sum (x m_i) \otimes n_i = \sum m \otimes (x n)$$

$$\begin{tikzcd}
\varphi: {}_KM \times {}_KN \arrow[r, "bilin"] \arrow[d] & {}_KL \\
(m,n) \arrow[r, mapsto] \arrow[d, mapsto] & \varphi(m,n) \\
m \otimes n \arrow[ur, dashed, "\exists! \bar{\varphi}"'] & \\
M \otimes N &
\end{tikzcd}$$

bilinearna

$$\varphi(xm, n) = x\varphi(m,n) = \varphi(m, xn)$$

baza

U, V vek. prostora $\quad U = \mathrm{span}\{e_i\} \quad V = \mathrm{span}\{f_j\}$

$$U \otimes V = \mathrm{span}\{e_i \otimes f_j\}$$

1. Zapis:

$$\left(\tfrac{1}{2},\tfrac{1}{2}\right)\otimes(1,1)+\left(\tfrac{1}{2},\tfrac{1}{2}\right)\otimes(0,1)+(-1,0)\otimes\left(\tfrac{1}{2},1\right)\in\mathbb{R}^2\otimes\mathbb{R}^2$$

kot enostavni tenzor $(u\otimes v)$

$$\left(\tfrac{1}{2},\tfrac{1}{2}\right)\otimes(1,0)+\left(\tfrac{1}{2},\tfrac{1}{2}\right)\otimes(0,1)+\left(\tfrac{1}{2},\tfrac{1}{2}\right)\otimes(0,1)+\tfrac{1}{2}(-1,0)\otimes(1,0)+$$
$$(-1,0)\otimes(0,1)=$$

$$=\left(0,\tfrac{1}{2}\right)\otimes(1,0)+(0,1)\otimes(0,1)=$$

$$=\underline{\underline{(0,1)\otimes\left(\tfrac{1}{2},1\right)}}$$

2.

Poišči primer vek. prost. U, V in tenzorja $x \in U \otimes V$, ki ni enostaven

$U, V = \mathbb{R}^2$

$U', V' = \mathbb{R}$

U', V' $\quad \sum (x_i \otimes y_i) = \sum x_i y_i (1 \otimes 1) = (1 \otimes \sum x_i y_i \, 1 \otimes 1)$ enostaven

$$\boxed{M \otimes_k k \cong M}$$

$$(1,0) \otimes (1,0) + (0,1) \otimes (0,1) = x \otimes y$$

$x = (x_1, x_2)$

$y = (y_1, y_2)$

$$x \otimes y = (x_1(1,0) + x_2(0,1)) \otimes (y_1(1,0) + y_2(0,1)) =$$

$$= x_1 y_1 (1,0) \otimes (1,0) + x_2 y_1 (0,1) \otimes (1,0) + x_1 y_2 (1,0) \otimes (0,1) + x_2 y_2 (0,1) \otimes (0,1)$$

$x_1 y_1 \neq 0 \Rightarrow x_1 y_1 \neq 0$

$x_2 y_1 = 0 \rightsquigarrow x_2 = 0$

$x_1 y_2 = 0$

$x_2 y_2 \neq 0 \rightsquigarrow x_2 \neq 0$ $\times$

$L^N = \mathrm{Hom}(N, L)$

Ali: velja $(L^N)^M \cong L^{N \times M}$

3. Pokaži ${}_k\mathrm{Hom}(M \otimes_k N, L) \cong \mathrm{Hom}(M, \mathrm{Hom}(N, L))$

$$\Phi : {}_k\mathrm{Hom}(M \otimes N, L) \longrightarrow \mathrm{Hom}(M, \mathrm{Hom}(N, L))$$

$$(\varphi : M \otimes N, L) \longmapsto (\psi : m \longmapsto (n \longmapsto \varphi(m \otimes n)))$$

$$\Psi : \mathrm{Hom}(M, \mathrm{Hom}(N, L)) \longrightarrow \mathrm{Hom}(M \otimes N, L)$$

$$\begin{tikzcd}
\psi \arrow[r, mapsto] & \varphi \arrow[d] \\
 & (m, n) \arrow[d, mapsto] \arrow[r, mapsto] & \psi(m)(n) \\
 & M \otimes n \arrow[ur, dashed, "\varphi"'] \arrow[ur, bend right]
\end{tikzcd}$$

Dokazati moramo še da je bilinearen

$\psi(m+m')(n) = (\psi(m) + \psi(m'))(n) = \psi(m)(n) + \psi(m')(n))$

ψ homomorfizem

$\psi(m)(n+n') = \psi(m)(n) + \psi(m)(n')$

$\psi(m)$ je homomorfizem

$\Phi \circ \Psi = id$

$(\Phi \circ \Psi)(\psi)(m)(n)$

$\Phi(\psi(m)(n)) = \Phi((m \otimes n) \mapsto \psi(m)(n)) = m \mapsto (n \mapsto \psi(m, n))(m)$

$\psi(m, n)$

$\forall$ parametrowe rozkłady

...

zatem iz te nalezą śled.

$$(\oplus M_i) \otimes N \cong \oplus (M_i \otimes N)$$

parábola sem gráfico

2025 1. izpit

K cel M, N tor. prosta

$M \otimes_K N$ neničelen

$$\boxed{K \otimes_K M = M}$$

Razširitev skalarjev

to je produkt dveh vek. p-ij. zato je produkt neničelen

$$(M \otimes_K N) \otimes_K F = (M \otimes_K F) \otimes_F (N \otimes_K F) \cong (M \otimes_K F) \otimes_F (F \otimes_K N)$$

$$\cong (M \otimes_K \underbrace{(F \otimes_F F)}_{\cong F}) \otimes_K N = (M \otimes_K N) \otimes_K F$$

$$\varphi : M \longrightarrow M \otimes_K F$$
$$m \longmapsto m \otimes 1$$
$$\ker \varphi = \operatorname{tors} M$$

Ker je M torzijsko prost ga lahko vložimo v $M \otimes_K F$

$$\left(M/_{\operatorname{tor} M} \hookrightarrow M \otimes_K F \right)$$

$$\Rightarrow M \otimes_K F = \{0\} \Rightarrow \ker \varphi = M = \{0\}$$

$$F^n \cong \frac{F[x]}{f_2(x)} \oplus \cdots \oplus \frac{F[x]}{f_n(x)}$$

Danes se mi ne da pisati